Project: Worked examples
Title: Beam formulas (single span)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Closed-form reactions, maximum moment and its position, maximum shear and maximum deflection for the standard single-span beam cases, as a quick check. Left end at x = 0; loads downward; sagging moments positive; deflection downward positive.

WHAT IT CALCULATES	RELATION
Formulas from Roark Table 8.1, in this sheet's own notation, for the chosen support and load case.	
Simply supported beam, point load at a (b = L − a), reaction and largest moment	R_A = Pb/L; M_max = Pab/L
Simply supported beam, point load at a, largest deflection when a ≥ b	δ_max = Pb(L² − b²)^(3/2)/(9√3 L EI) at x = √((L² − b²)/3)
Propped cantilever, uniform load, reaction and moments	R_B = 3wL/8; M_A = −wL²/8; sagging M_max = 9wL²/128 at x = 5L/8

Basis: Shear, moment, slope and deflection formulas for elastic straight beams[1](Ch. 8, Table 8.1).

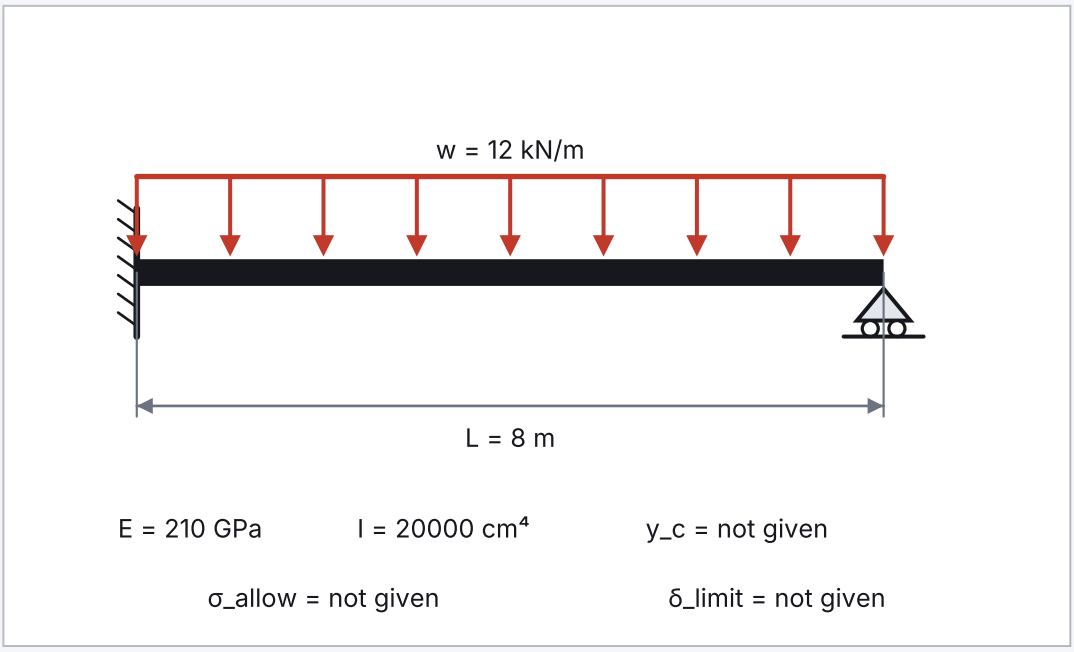


Figure 1 — Propped cantilever, fixed left and pinned right, uniformly distributed load w. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
case	Case	propped cantilever (fixed left, pinned right), UDL		assumption: Fixed at the wall, propped at the free end.	any
L	Span	8	m	drawing: DRG-CA-033 rev A	0.1 … 100 m
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
I	Second moment of area	20000	cm^4	reference: Fabricated section, S1 sheet	1 … 100000000 cm^4
w	Distributed load (downward)	12	kN/m	spec: Load schedule LS-03	0 … 10000 kN/m

3. Assumptions

A1. Prismatic, linear elastic, Euler–Bernoulli beam; small deflections. (Roark Table 8.1 conditions.)
A2. A cantilever is fixed at the left end (x = 0) and free at the right. (Case convention.)
A3. case (Case) is taken as propped cantilever (fixed left, pinned right), UDL. (Fixed at the wall, propped at the free end.)

4. Working

[5] EI = E * I
= 210 GPa × 20000 cm^4
= 42000 $kN·m^2$
Flexural rigidity

[7] R_A = 5 * w * L / 8 [1](8.1 2c)
= 5 × 12 kN/m × 8 m / 8
= 60 kN
Fixed-end reaction

[8] R_B = 3 * w * L / 8
= 3 × 12 kN/m × 8 m / 8
= 36 kN
Propped-end reaction

[9] M_min = -w * L^2 / 8
= −12 kN/m × (8 m)^2 / 8
= -96 kN·m
Fixed-end (hogging) moment

[10] M_max = 9 * w * L^2 / 128
= 9 × 12 kN/m × (8 m)^2 / 128
= 54 kN·m
Maximum sagging moment

[11] x_M = 5 * L / 8
= 5 × 8 m / 8
= 5 m
Position of the maximum sagging moment

[12] V_max = 5 * w * L / 8
= 5 × 12 kN/m × 8 m / 8
= 60 kN
Maximum shear

```
[13] x_delta = L * (1 + sqrt(33)) / 16
= 8 m × (1 + sqrt(33)) / 16
= 3.3723 m
```
Position of the maximum deflection

```
[14] delta_max = w * L^4 / (185 * EI)
= 12 kN/m × (8 m)^4 / (185 × 42000 kN·m²)
= 6.3259 mm
```
Maximum deflection

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	60 kN	Left reaction
R_B	36 kN	Right reaction
M_max	54 kN·m	Maximum sagging moment
x_M	5 m	Position of M_max
M_min	-96 kN·m	Minimum (hogging) moment
V_max	60 kN	Maximum shear
delta_max	6.3259 mm	Maximum deflection
x_delta	3.3723 m	Position of δ_max

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

Template struct.beam-formulas@1.0.1 Content hash 9ce95307c4fbf44b Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.